

SENTINEL

Autonomous AI Financial Crime Investigation & Threat Intelligence Platform

Hackathon project blueprint — designed for a high-impact 36-hour build.

Winning thesis: Don't build another chatbot or simple fraud classifier. Sentinel takes evidence, reconstructs the attack, connects entities, traces the synthetic money flow, explains its findings, and produces an investigation report.

1. Executive Summary

Sentinel is an AI-powered investigation platform for phishing, payment fraud, mule-account activity and related cyber-financial attacks. A user or investigator can upload screenshots, URLs, QR codes, transaction records, emails or PDFs. Sentinel extracts entities and indicators, runs threat and financial analysis, builds a relationship graph, produces an explainable risk score, and generates recommended next actions. A blockchain-backed evidence hash provides tamper-evident integrity for the investigation record.

2. The Problem

- Most fraud tools stop at detection: fraud / not fraud.
- Investigators often need to correlate scattered evidence across messages, URLs, accounts, transactions and wallets.
- A high-risk score without evidence is difficult to trust or act upon.
- Evidence can be difficult to preserve and verify across an investigation.

3. The Solution

Core promise: “Give Sentinel the evidence. It will reconstruct the attack.”

The MVP follows this pipeline:

Upload evidence → multimodal extraction → threat analysis → financial analysis → entity resolution → graph construction → agentic investigation → explainable risk score → actionable report → evidence integrity verification.

4. The Hackathon Showstopper Demo

Step	Demo action	What judges see
1	Upload a synthetic scam screenshot + transaction record	Evidence appears in the case workspace
2	AI extracts phone, URL, UPI ID, amount and suspicious details	Spotted entities appear automatically
3	Threat agent analyzes the URL/domain	Risk factors and confidence appear
4	Financial agent analyzes the transaction	Anomaly indicators appear
5	Graph agent connects entities	Interactive victim → phone → domain → UPI → account → wallet graph
6	Investigation agent reasons over the evidence	Attack narrative + evidence-backed findings
7	Generate report	Professional investigation summary + recommended actions
8	Verify evidence	SHA-256 hash + blockchain verification status

5. Product Features

- Multimodal Evidence Intake** — Screenshot, image, PDF, email text, URL, QR code and transaction data.
- AI Evidence Extraction** — Extract entities, indicators, amounts, timestamps, claims and suspicious patterns.
- Threat Intelligence** — Score domains, URLs and other indicators using demo data and optional external feeds.
- Financial Analysis** — Detect unusual amount, velocity, beneficiary and transaction-chain patterns.
- Investigation Graph** — Represent relationships between people, phones, domains, UPI IDs, accounts, transactions and wallets.
- Explainable Risk** — Show exactly which evidence contributed to the score instead of returning an unexplained number.
- Automated Report** — Generate a case summary, findings, confidence and recommended actions.
- Evidence Integrity** — Hash uploaded evidence and record the hash on a local/test blockchain.

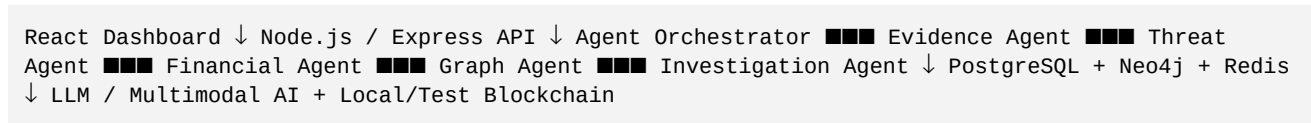
6. AI Agent Architecture

Agent	Responsibility
Evidence Agent	OCR/multimodal extraction; normalize raw evidence into structured entities and indicators.
Threat Intelligence Agent	Analyze URLs, domains, IPs and phishing indicators.
Financial Intelligence Agent	Analyze transaction anomalies and synthetic money-flow patterns.
Graph Agent	Create and query entity relationships; identify connected suspicious entities.
Investigation Agent	Combine findings, reason over evidence, explain the attack and produce the final report.

7. Recommended Architecture

React + Vite frontend → Node.js/Express API → agent orchestration → PostgreSQL/Prisma for case data → Neo4j for investigation relationships → Redis for caching/job state → Gemini or another multimodal LLM API → local Hardhat blockchain for evidence-hash verification.

Suggested architecture flow:



8. Technology Stack & Cost

Component	Recommended choice	Hackathon cost
Frontend	React + Vite + Tailwind	Free
Backend	Node.js + Express	Free
Database	PostgreSQL + Prisma	Free locally
Graph DB	Neo4j Community	Free locally
Cache	Redis	Free locally
AI orchestration	LangChain / LangGraph	Open source
OCR	Tesseract	Free

Component	Recommended choice	Hackathon cost
Blockchain	Hardhat + Solidity	Free locally
Containerization	Docker	Free for this use
LLM	Gemini or equivalent API	Potential API cost / available quota
Cloud	Optional	Can be 0 using local/demo deployment or a suitable free tier

Practical budget target: 0–2,000. The main variable is LLM/API usage; avoid paid cloud infrastructure unless the event requires a public deployment.

9. Database Model

PostgreSQL stores application and case data. Neo4j stores relationships that are better represented as a graph.

PostgreSQL entities: User, Case, Evidence, Finding, Transaction, Report, Action, EvidenceHash.
Neo4j nodes: Person, Phone, Email, Domain, URL, UPI, BankAccount, Transaction, Wallet. Neo4j
relationships: OWNS, USES, SENT, PAID_TO, LINKED_TO, FUNDS, ASSOCIATED_WITH.

10. Example Investigation

Input: A synthetic SMS claims that a bank account will be suspended and contains a suspicious URL. A transaction screenshot shows a ■48,500 payment to a new beneficiary.

Expected output: Sentinel identifies impersonation language, a suspicious domain, an anomalous transaction and connected synthetic entities. The graph shows a possible victim → phone → phishing domain → UPI → account → mule account → crypto wallet chain. The system reports a high risk score with evidence-backed reasons rather than pretending the model has certainty.

11. Important Safety / Data Rule

Use only synthetic or publicly permitted demo data during the hackathon. Do not upload real customers' banking details, personal documents, private phone numbers or financial records. Clearly label the money-flow and threat-intelligence datasets as simulated when they are simulated.

12. 36-Hour Execution Plan

Time	Goal
0–4 h	Finalize story, architecture, database schema, UI wireframe and synthetic demo dataset.
4–10 h	Build React dashboard, case creation, evidence upload and backend API.
10–18 h	Implement AI extraction, structured outputs and threat/financial agents.
18–25 h	Implement Neo4j graph, investigation flow and graph visualization.
25–30 h	Generate explainable report, evidence hashing and local blockchain verification.
30–34 h	Polish UI, animations, loading states, error handling and demo data.
34–36 h	Rehearse pitch, test demo offline/fallback path and prepare judging answers.

13. What NOT to Build

- A generic PDF chatbot or RAG assistant.
- A simple “fraud / not fraud” classifier.
- A basic expense tracker or payment application.
- Blockchain added only for decoration.
- A giant feature list with no reliable end-to-end demo.

14. Judge-Focused Differentiators

- **Novelty:** investigation rather than detection alone.
- **Technical depth:** multimodal AI + agents + graph database + financial analytics + blockchain evidence integrity.
- **Demo impact:** one uploaded screenshot can trigger a complete investigation flow.

- **Explainability:** every major conclusion is connected to evidence.
- **Business potential:** the same workflow can be adapted for banks, fintechs, enterprises and investigation teams.
- **Reliability:** use a controlled synthetic dataset and deterministic fallback paths so the demo does not depend entirely on external services.

15. Final Pitch

Sentinel is an autonomous AI investigation platform for cyber-financial crime. Instead of simply flagging a suspicious transaction or phishing message, Sentinel combines multimodal evidence analysis, threat intelligence, financial anomaly detection and graph reasoning to reconstruct how an attack happened, connect the entities involved, explain why it is suspicious, and generate an actionable investigation report. Blockchain-backed evidence hashes provide an additional integrity layer.

Tagline: “One screenshot. Full investigation.”